

Assignment 1

Part 1.

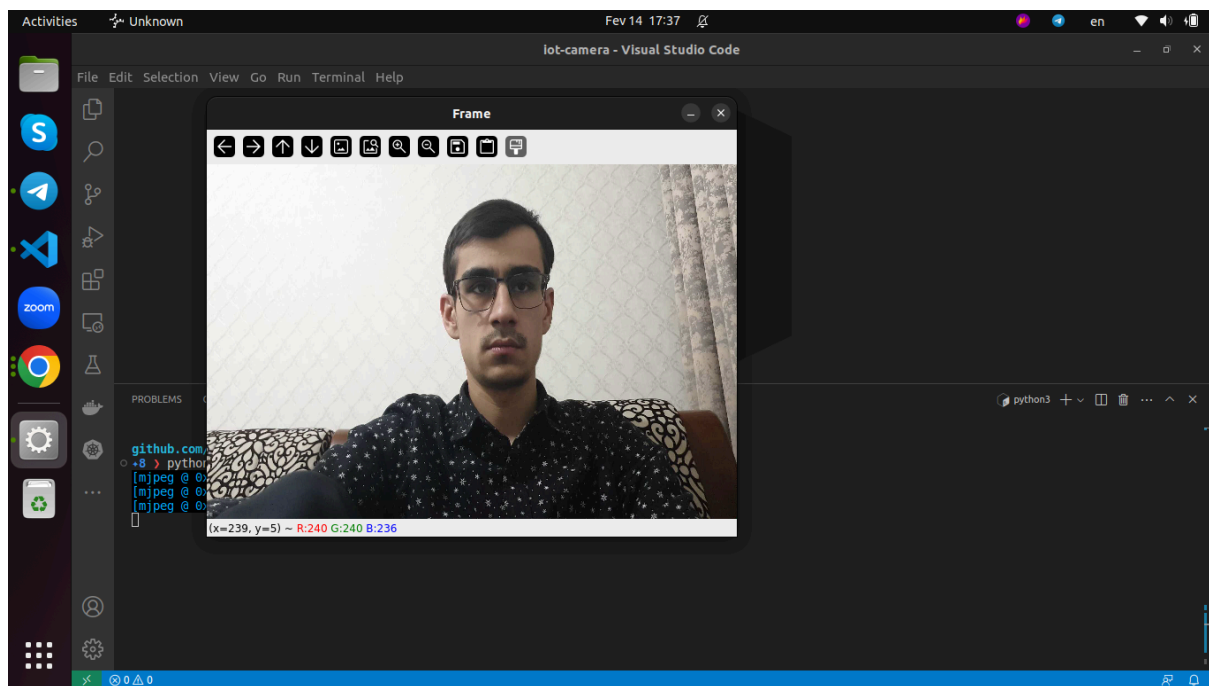
Access phone's camera with python

Take **2 screenshots**

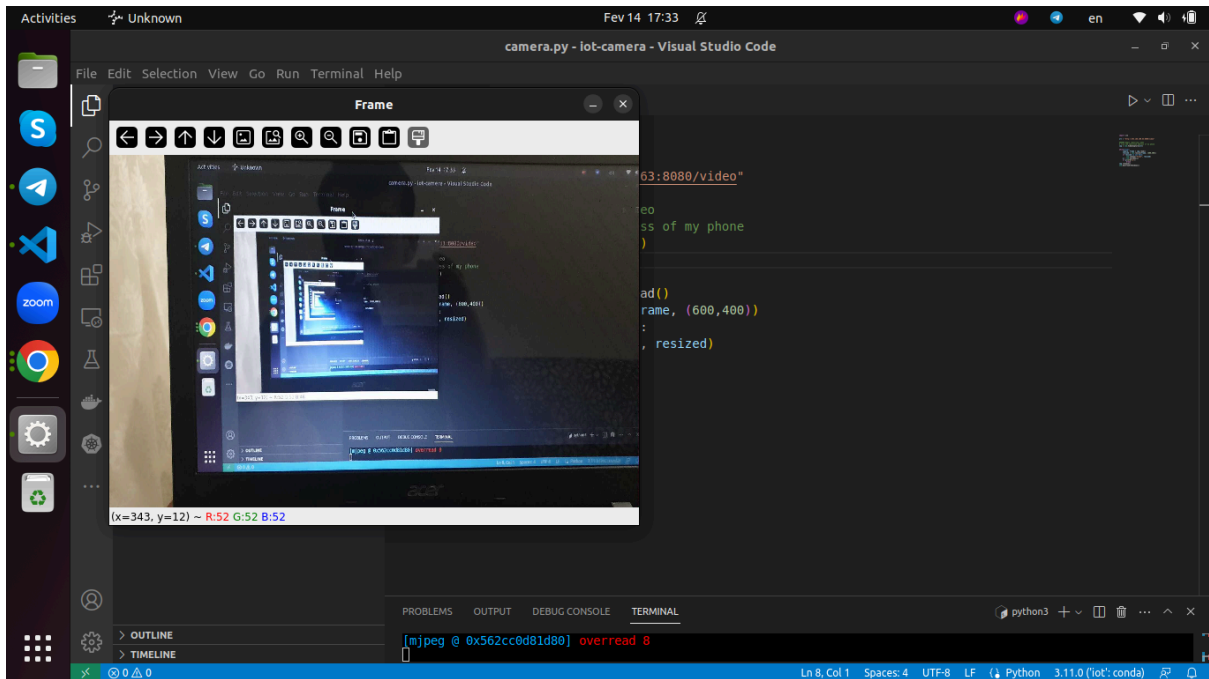
- when you yourself and your desktop open applications(vscode) are visible like this one
- While pointing the camera to your desktop so that there is infinite window loop visible

In your screenshots, you must include the **whole** desktop, not just the picture part just like I did in the following two examples.

Screenshot 1



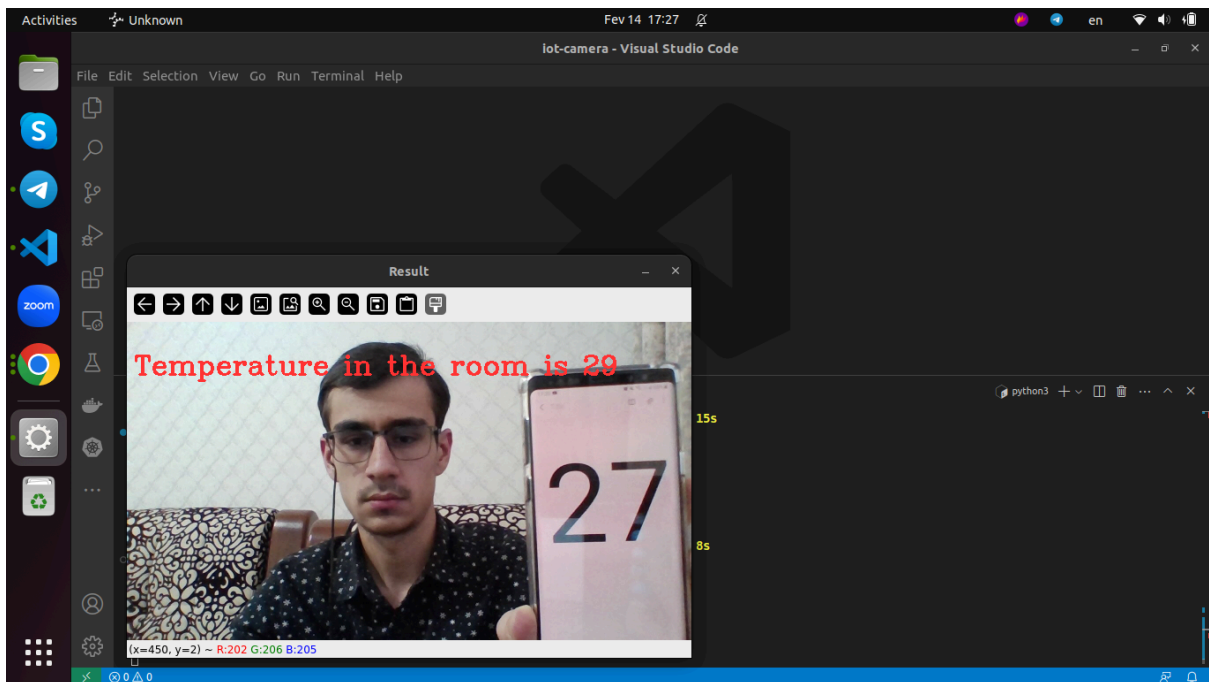
Screenshot 2



Part 2.

This time, we will mimic detecting room temperature with a remote IoT device. As we do not have sensor devices, you should show a number to the camera, which stands for the temperature to the camera and detect this number in your code and print out the detected number in the video.

The result should be like the following
Screenshot 3

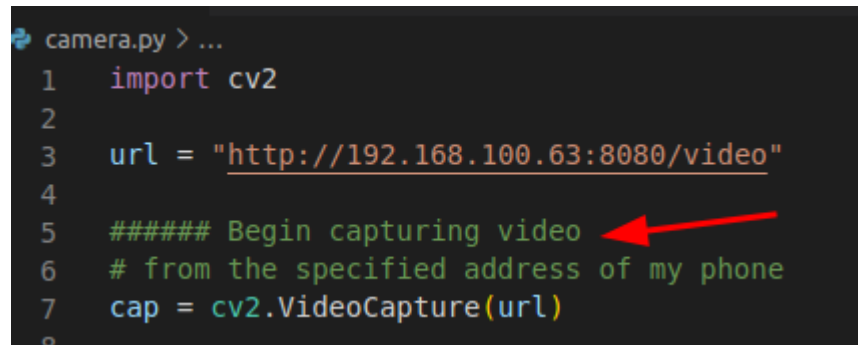


You must include a screenshot like this where you and the number are visible.
As you can see my detection was not perfect. It is actually 27 but shows 29. It is still okay.

CODE

Comment important lines, functions in your code correspondingly with its function.

E.g.



```
camera.py > ...
1  import cv2
2
3  url = "http://192.168.100.63:8080/video"
4
5  ##### Begin capturing video
6  # from the specified address of my phone
7  cap = cv2.VideoCapture(url)
8
```

You must have at least two comments

- For video capture code snippet
- For digit detection code snippet

What to submit

All your code files together with screenshots and a requirements.txt file.

Name your code files part_1.py and part_2.py

Name your screenshots screenshot_1.png, screenshot_2.png, screenshot_3.png

Zip them together and name that zip as your ID.

Grading Criteria

Part 1	Video capture code	2
	Code pointed out with comment	1

	Result screenshot 1	2
	Result screenshot 2	2
Part 2	Digit Detection Code	2
	Code pointed out with a comment	1
	Result screenshot	2
Not following naming conventions for files, wrong submission format, etc.		-3
Max		12

WARNING!!!

Same screenshot or any other elements of cheating found in your submission -> zero.

References

Capture video with camera

▶ [How to Access Mobile Phone Camera using OpenCV | Machine Learning | Data Magic](#)

It turns out the above video is not available anymore. Please use the following video instead.

▶ [How to use Phones Camera as OpenCV Webcam for Face Detection](#)

Text Detection OCR with python

▶ [Text Detection with OpenCV in Python | OCR using Tesseract \(2020\)](#)

Recognize Digits with python

▶ [Digit Recognition with Webcam | OpenCV](#)